

2. Champagne Tower

Aim

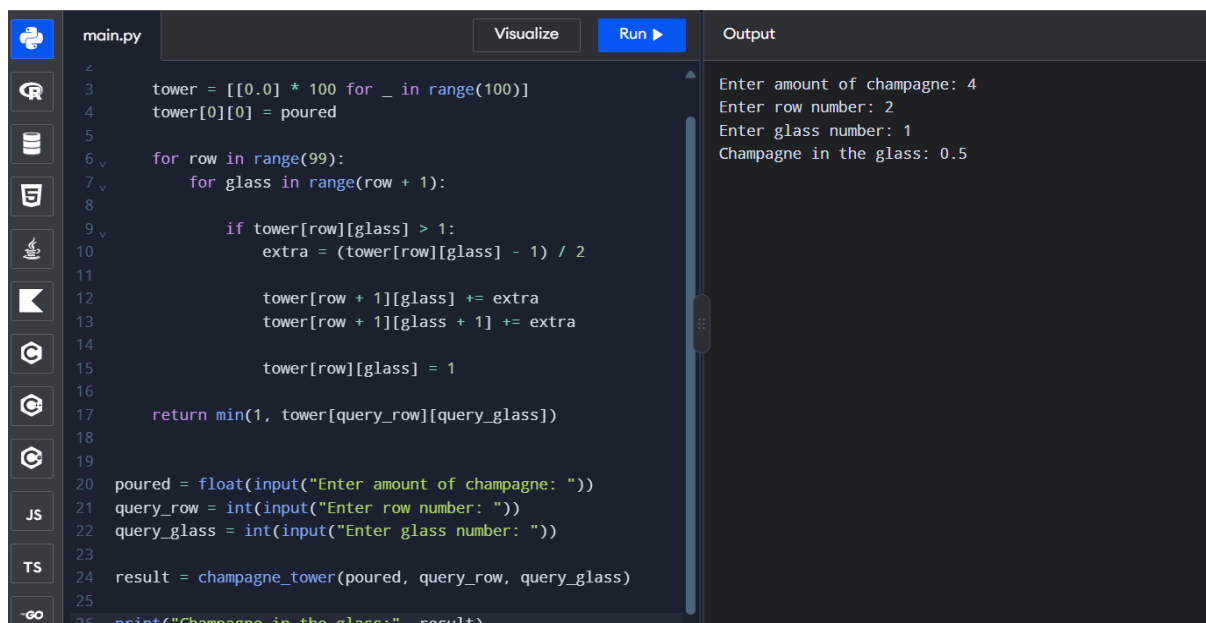
To calculate the quantity of champagne available in a specified glass of the champagne tower after pouring a given amount of champagne.

Algorithm

1. Accept the total quantity of champagne poured into the top glass.
2. Initialize the champagne tower with the top glass containing the poured amount.
3. Check each glass to determine whether it contains more than 1 unit of champagne.
4. If a glass overflows, divide the excess champagne equally between the two glasses in the next row.
5. Continue distributing the excess champagne row by row.
6. Identify the required glass using its row and position.
7. Limit the quantity in each glass to a maximum of 1 unit.
8. Display the amount of champagne present in the specified glass.

Procedure

1. Open Python IDLE, VS Code, or an online Python compiler.
2. Create a new Python file and enter the given program.
3. Execute the program.
4. Provide the total amount of champagne poured when prompted.
5. Enter the row number and glass position to be checked.
6. The program distributes the excess champagne among the lower glasses.
7. It calculates the quantity present in the selected glass.
8. Display and record the obtained result.



The screenshot shows a Python IDE with a file named 'main.py'. The code implements the Champagne Tower algorithm. It initializes a 100x100 grid representing the tower, where the top glass contains the poured amount. The algorithm iterates through each glass, checking if it contains more than 1 unit. If it does, the excess is divided equally between the two glasses in the next row. The process continues until the excess is distributed. Finally, the amount in the specified glass is returned, limited to a maximum of 1 unit. The output shows the result for 4 units poured, row 2, glass 1, which is 0.5.

```
main.py Visualize Run ▶ Output
1 tower = [[0.0] * 100 for _ in range(100)]
2 tower[0][0] = poured
3
4
5
6 for row in range(99):
7     for glass in range(row + 1):
8
9         if tower[row][glass] > 1:
10            extra = (tower[row][glass] - 1) / 2
11
12            tower[row + 1][glass] += extra
13            tower[row + 1][glass + 1] += extra
14
15            tower[row][glass] = 1
16
17
18     return min(1, tower[query_row][query_glass])
19
20 poured = float(input("Enter amount of champagne: "))
21 query_row = int(input("Enter row number: "))
22 query_glass = int(input("Enter glass number: "))
23
24 result = champagne_tower(poured, query_row, query_glass)
25
26 print("Champagne in the glass:", result)
```

Enter amount of champagne: 4
Enter row number: 2
Enter glass number: 1
Champagne in the glass: 0.5

Result

Thus, the amount of champagne present in the specified glass of the champagne tower was successfully calculated using Python.